

We use the symbol $\subset$ to denote the subset relation without enforcing irreflexivity.

Separating the Dynamics from the initialisation of a stochastic process

Fix a domain for stochastic processes $D \subset \mathbb{R}^N$ and a time index set $I \subset [0, \infty)$ (so I can be discrete as well, $I = \mathbb{N}$).

On a measurable space $(\Omega, \mathcal{F})$ a **stochastic process** is an I -indexed sequence of random variables with values in D , i.e. a map $X : I \times \Omega \rightarrow D$ where $X(t, \cdot)$ is measurable at each fixed time $t \in I$.

This setup already fixes a starting position — the random variable X_0 with its law $\mathbb{P}_{X_0}$. So there are two aspects to this definition that we need to separate conceptually.

Recall that a Markov chain is a stochastic process X satisfying the simple Markov property

$$\forall t_1 < \dots < t_{k+1} \in I : \mathbb{P}_{X_{t_{k+1}} | X_{t_1}, \dots, X_{t_k}} = \mathbb{P}_{X_{t_{k+1}} | X_{t_k}}.$$

A Markov process induces a kernel (*transition matrix* if D is discrete)

$$\begin{aligned} & \text{for any } t \in I, \quad P_t : D \times \mathcal{F} \rightarrow [0, 1] \\ & P_t(x, A) := P(X_t \in A \mid X_0 = x), \end{aligned}$$

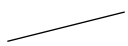
but a kernel on its own does not determine a stochastic process, not at least until a *starting point* (or distribution) is specified — different processes started at different points can share the same kernel (dynamics). In discrete time and space, it is known that a transition matrix with a choice of a starting point induces a Markov chain.

A stochastic differential equation of the form $dX_t = b(t, X)dt + \sigma(t, X)dw_t$, $X_0 = x_0$ may be solved by a stochastic process. While the equation + the starting point are what determine the solution, and we are used to saying that the expression $dX_t = b(t, X)dt + \sigma(t, X)dW_t$ is meaningless on its own, it is useful to treat the two as conceptually standalone pieces of the definition — akin to how the Markov kernel is a standalone object paired with each Markov chain. After all, the same b, σ induce different solutions for different starting points x_0 .

Markov chain action on the space of distributions

The sampling problem

As described in [Andrea], we consider 3 classes of sampling problems depending on what is known:

Task: produce a sample $x \sim \mu$ where		
	we know	but do not know
Case I.	the distribution μ	
Case II.	a family of distributions and a sample sequence $(\mu_\theta)_\theta$ and $(x_i)_i \stackrel{\perp}{\sim} \mu_\theta$	the parameter θ , and hence which μ_θ
Case III.	a sample sequence only $(x_i)_i \stackrel{\perp}{\sim} \mu$	the distribution μ

We will work in case I. for now and assume we know the distribution μ . Diffusion based sampling is the approach of approximating μ by a sequence $(\nu_t)_t$

Chapman Kolmogorov

Here I follow [Zaragoza] Assume a reference measure exists with respect to which the laws $\mathbb{P}_{Y_i}, i \in I$ have densities p_i . I assume it is the Lebesgue measure that works, but maybe this discussion works for other reference measures too?

Kolmogorov equations

Denote $p_{i|j}(y | z) = p_{ij}(y, z)/p_j(z)$ the conditional density.

The Chapman-Kolmogorov equation states that the probability density of getting from y_1 to y_3 is

$$p_{3|1}(y_3 | y_1) = \int p_{3|2}(y_3 | y) p_{2|1}(y | y_1) dy$$

The meaning is clear.

The Master equation is

$$\begin{aligned} \partial_t p_{t|t_0}(y|y_0) = & \int \left(W_{y|y'}^{\text{in}} \cdot p_{t|t_0}(y'|y_0) \right. \\ & \left. - W_{y'|y}^{\text{out}} \cdot p_{t|t_0}(y|y_0) \right) dy' \end{aligned}$$

Now expand the right hand side as a function of $r := y' - y$ around $r = 0$.

For diffusions we can drop higher order terms and get Fokker-Planck

$$\begin{aligned} \partial_t p_t(y) = & -\partial_y(a^{(1)}(y, t)p_t(y)) \\ & + \frac{1}{2}\partial_y^2(a^{(2)}(y, t)p_t(y)) \end{aligned}$$

In infinitesimal form this requires a bit more machinery. We define the infinitesimal generator of X by

$$\begin{aligned} \mathcal{L}\varphi = & \left(x \mapsto \lim_{h \rightarrow 0} \frac{P_h \varphi - \varphi}{h}(x) \right) \\ = & \left. \frac{dP_t}{dt} \right|_{t=0}(\varphi) \end{aligned}$$

which tells us by how much a test function φ (thought as a distribution density) is pushed by X_t in an infinitesimal time. For a time-homogeneous $dX_t = b(X_t)dt + \sigma(X_t)dW_t$ this is¹

$$\mathcal{L} = b^i \partial_i + \frac{1}{2} a^{ij} \partial_i \partial_j, \quad a := \sigma \sigma^\top.$$

Then $u(t, x) := (P_t \varphi)(x)$ satisfies $\partial_t u = \mathcal{L}u$

Next we define the adjoint $\mathcal{L}^*$ and forward Kolmogorov is

$$\begin{aligned} \partial_t p_t(x, y) = & \mathcal{L}_y^+ p_t(x, y) \\ \mathcal{L}^+ \varphi = & \frac{1}{2} \partial_i \partial_j a^{ij} \varphi - \partial_i b^i \cdot \varphi \end{aligned}$$

where $p_t(x, y)$ is transition density from x to y in time t (for time-homogeneous systems it doesn't matter at which time t_0 we start, only the time difference t).

Then if

$$\begin{aligned} \rho(x) = & \frac{d\mathbb{P}_{X_0}}{dx}(x), \\ \rho(t, y) := & \int p_t(x, y) \rho(x) dx, \end{aligned}$$

Fokker-Planck is

$$\partial_t \rho(t, y) = \mathcal{L}_y^+ \rho(t, y).$$

¹using Einstein summation

Ornstein-Uhlenbeck

Let $x \sim \mu$ such that $x \perp W$.

$$\begin{aligned}dZ_t &= -Z_t dt + \sqrt{2} dW_t \\ Z_0 &= x\end{aligned}$$

From $e^{-t} d e^t Z_t = dZ + Z dt$ we find the solution is

$$\begin{aligned}Z_t &= e^{-t} x + \sqrt{2} e^{-t} \int_0^t e^s dW_s \\ &= e^{-t} x + \sqrt{1 - e^{-2t}} G_t, \quad (x, G_t) \sim \mu \otimes \mathcal{N}(0, 1)\end{aligned}$$

where we rescaled the stochastic integral to

$$G_t := (1/2(e^{2t} - 1))^{-1/2} \int_0^t e^s dW_s$$

to get a unit normal² $G_t \sim \mathcal{N}(0, 1)$.

² $\int_0^t e^s dW_s$ is normal of variance $V = \int_0^t e^{2s} ds = \frac{1}{2}(e^{2t} - 1)$, thus $G_t := V^{-1/2} \int_0^t e^s dW_s$ is normal of variance 1.

Wiener process invariant measure

We are looking for an invariant measure of the Wiener process W_t , i.e. for a solution μ of the measure equation

$$\mu \cdot W_t = \mu.$$

We are looking only for *finite* measures $\mu : \mathcal{B}(\mathbb{R}) \rightarrow [0, \infty)$, so that we are allowed to subtract $\mu(A)$ as we wish for any Borel A .

We apply both sides at the (measurable) set $(-\infty, a)$ for $a \in \mathbb{R}$. Evaluate

$$(\mu \cdot W_t)(-\infty, a) = \mathbb{P}(W_t^\mu < a) = \int \Phi_t(a - x) d\mu(x).$$

Now write $\mu(-\infty, a) = \int 1_{(-\infty, a)}(x) d\mu(x) = \int 1_{(0, \infty)}(a - x) d\mu(x)$.

Then $\mu \cdot W_t = \mu$ at $A = (-\infty, a)$ reads, after rearranging,

$$\int (\Phi_t - 1_{(0, \infty)})(a - x) d\mu(x) = 0. \quad (1)$$

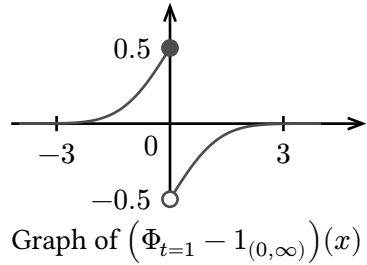
Curiously this is the statement that the convolution of the function $\Phi_t - 1_{(0, \infty)}$ with the measure μ vanishes at the point a .

So $\mu \cdot W_t = \mu$ is equivalently³ written as

$$(\Phi_t - 1_{(0, \infty)}) * \mu = 0$$

as an equality of functions $\mathbb{R} \rightarrow \mathbb{R}$.

So an invariant measure of W_t has to satisfy this convolutional equation.



We look now at that integral separately on $(-\infty, a)$ and⁴ $[a, \infty)$:

³because $(-\infty, a), a \in \mathbb{R}$ generate $\mathcal{B}(\mathbb{R})$, we are not losing information when evaluating $\mu \cdot W_t = \mu$ at $(-\infty, a)$.

⁴we denote the inclusion of the point $x = a$ in the integral boundary by $\underline{a}$. So $\int_a^b$ denotes the integral over (a, b) , and $\int_{\underline{a}}^b$ denotes the integral over $[a, b)$.

$$\begin{aligned}
\forall a : 0 &\stackrel{(1)}{=} \int (\Phi_t - 1_{(0,\infty)})(a-x) d\mu(x) \\
&= \int_{-\infty}^a (\Phi_t(a-x) - 1) d\mu(x) + \int_a^{\infty} \Phi_t(a-x) d\mu(x).
\end{aligned}$$

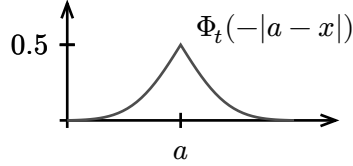
Up to this point we haven't used any knowledge about Φ_t so this equality applies to the law of any process. Now we kill the (-1) in the first term by applying $\Phi_t(y) - 1 = -\Phi_t(-y)$, which gets us the equality

$$\forall a : \int_{-\infty}^a \Phi_t(-(a-x)) d\mu(x) = \int_a^{\infty} \Phi_t(a-x) d\mu(x),$$

and if we notice that the argument to the Φ_t is $-|a-x|$ in both domains:

$$\forall a : \int_{-\infty}^a \Phi_t(-|a-x|) d\mu(x) = \int_a^{\infty} \Phi_t(-|a-x|) d\mu(x). \quad (2)$$

That is, any W_t -invariant measure μ must give the same integral of Φ on both sides of any real a . This kind of super-symmetry indicates that μ should be somehow similarly behaved everywhere.



The graph of the integrands in eq. 2

We can extract $\mu(\text{interval})$ expressions from the integrals above by rewriting $\Phi(-|a-x|) = 1 - \Phi(|a-x|)$ in eq. 2:

$$\forall a : \mu(-\infty, a) - m_t^a(-\infty, a) = \mu[a, \infty) - m_t^a[a, \infty),$$

where $m_t^a(A) := \int_A \Phi_t(|a-x|) d\mu(x)$. Or, more succinctly,

$$\forall a : (\mu - m_t^a)(-\infty, a) = (\mu - m_t^a)[a, \infty). \quad (3)$$

Having an expression relating μ on each pair of intervals $(-\infty, a), [a, \infty)$, it is natural to express an interval $[a, b)$ in two ways and derive an expression for $\mu[a, b)$.

Pick reals $a < b$ and instantiate eq. 3 with a and b , and in both cases split the measures (integrals) at the points a and b :

$$\begin{aligned}(\mu - m_t^a)(-\infty, a) &= (\mu - m_t^a)([a, b] \cup [b, \infty)) \\ (\mu - m_t^b)((-\infty, a) \cup [a, b)) &= (\mu - m_t^b)[b, \infty).\end{aligned}$$

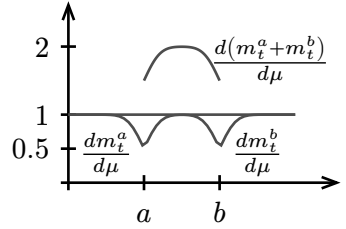
Now subtract one equation from the other⁵ and solve for $\mu[a, b]$:

$$2\mu[a, b] = (m_t^b - m_t^a)(-\infty, a) + (m_t^b + m_t^a)[a, b] + (m_t^a - m_t^b)[b, \infty)$$

Now we bound each term:

- on $(-\infty, a)$ we have $m_t^b \geq m_t^a$,
- on $[b, \infty)$ we have $m_t^a \geq m_t^b$,
- on $[a, b]$ we have $m_t^a + m_t^b \geq c_{ab}\mu$ for a constant

$$c_{ab} = \inf_{a \leq t < b} (\Phi(|a - x|) + \Phi(|b - x|)).$$



So we bound $2\mu(a, b) \geq 0 + c\mu(a, b) + 0 = \mu(a, b)$, forcing $\mu(a, b) = 0$.

⁵to keep the $\mu[a, b]$ terms and kill the $\mu((-\infty, a) \cup [b, \infty))$ terms.

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Put $A = (-\infty, a)$. Write each of $\mu, \mu \cdot W_t$ as a μ -integral of a function of argument $(a - x)$:

$$\begin{array}{l|l} (\mu \cdot W_t)(A) = \mathbb{P}_{W_t}^\mu(-\infty, a) & \mu(A) = \mu(-\infty, a) \\ = \int \mathbb{P}_{W_t}^x(-\infty, a) \, d\mu(x) & = \int 1_{(-\infty, a)}(x) \, d\mu(x) \\ = \int \Phi_t(a - x) \, d\mu(x) & = \int 1_{(0, \infty)}(a - x) \, d\mu(x) \end{array}$$

Evaluate the difference $\mu - \mu \cdot W_t$:

$$(\mu - \mu \cdot W_t)(A) = \int (1_{(0, \infty)} - \Phi_t)(a - x) \, d\mu(x)$$

(Partition the integral on $(-\infty, a)$ and $[a, \infty)$)

$$= \int_{-\infty}^a (1 - \Phi_t(a - x)) \, d\mu(x) - \int_a^\infty \Phi_t(a - x) \, d\mu(x)$$

(Apply $1 - \Phi_t(y) = \Phi_t(-y)$ on the left)

$$\begin{aligned} &= \int_{-\infty}^a \Phi_t(-(a - x)) \, d\mu(x) - \int_a^\infty \Phi_t(a - x) \, d\mu(x) \\ &= \int_{-\infty}^a \Phi_t(-|a - x|) \, d\mu(x) - \int_a^\infty \Phi_t(-|a - x|) \, d\mu(x) \\ &= m_t^a(-\infty, a) - m_t^a[a, \infty) \end{aligned}$$

where $m_t^a(A) := \int_A \Phi_t(-|a - x|) \, d\mu(x)$.